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Effects of Mn, Ti and V on the microstructure and properties of AlCrFeCoNiCu high entropy alloy

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ABSTRACT

Mn, Ti and V were added in an equal-molar ratio to the AlCrFeCoNiCu alloy and the effects of these added elements on microstructure and properties of AlCrFeCoNiCu high entropy alloy were studied. The results indicate that the AlCrFeCoNiCuMn alloy shows almost the same microstructure as in AlCrFeCoNiCu alloy except for a long-strip Cr-enriched phase. The additions of Ti change the AlCrFeCoNiCu alloy from a dendrite structure to eutectic-cell one. Hence two phases in the eutectic cells are detected as one Al, Ti, Co, Ni enriched phase and another Cr, Fe enriched phase. V added alloy is also a dendrite structure, but V additions change the morphology of dendritic area from modulated plates to ellipsoidal particles. The addition of V into the AlCrFeCoNiCu alloy shows the best strengthening effect and the lowest decrease in the ultimate strain in our experiments.

1. Introduction

Up to present, traditional alloys, including Fe-, Al-, Cu-, Mg-, Ti-, Ni-, TiAl-, NiAl- and FeAl-based alloys, are typically composed of one or two principal elements which with only minor additions of other elements to modify their properties [1–4]. In this case, multi-principal elements alloys have been limited by the traditional metallurgical understanding and engineering practice [5,6].

However, this paradigm has been broken by high-entropy alloys defined as those containing at least five principal elements with the concentration between 5 and 35 at.% for each element. Yeh et al. considered that these alloys possess higher mixing entropy, therefore favors the formation of multi-element solid solution, as opposed to the inferred complex structures consisting of many intermetallic compounds [7–9]. By proper composition design, high entropy alloys can obtain promising properties in hardness, wear resistance, oxidation resistance and corrosion resistance [10–16].

Many high entropy alloy systems have been exploited in the past decade. AlCrFeCoNiCu alloy is a widely studied high strength high entropy alloy [9,17–19]. In this paper, Mn, Ti and V were added into AlCrFeCoNiCu alloy to form septenary equimolar high entropy alloys. And their effects on structures and properties of AlCrFeCoNi high entropy alloy were investigated.

2. Materials and experimental details

The AlCrFeCoNiCu, AlCrFeCoNiCuMn, AlCrFeCoNiCuTi and AlCrFeCoNiCuV (nominal compositions is equimolar) alloy ingots were prepared by vacuum arc melting the mixtures of pure metals with purities of better than 99.9 mass%, in a Ti-gettered high-purity argon atmosphere. The ingots were remelted four times to improve homogeneity. Alloy rods were cast from the ingots into a copper molds with 10 mm diameter.

The structural features of as-cast alloy samples were examined with a D/MAX-rB X-ray diffractometer (XRD) with Cu Kα radiation ($\lambda = 0.154$ nm). The microstructure was characterized using a Hitachi S-4700 scanning electron microscope (SEM) equipped with energy dispersive spectrometry (EDS). Transmission electron microscope (TEM) investigations were performed with a FEI Tecnail G20 with 200 kV acceleration voltage. The room temperature mechanical properties were evaluated by uniaxial compression tests on samples of Φ 3 mm × 5 mm in an Instron 5500 testing system at a loading speed of 0.5 mm/min. Three compression tests were performed to obtain the average value.

3. Results and discussion

3.1. XRD analysis

Fig. 1 shows the XRD patterns of the as-cast high entropy alloys. AlCrFeCoNiCu is consisted of a BCC phase and a FCC phases. AlCrFeCoNiCuMn alloy is consisted of a BCC phase, a FCC phase and an unknown phase. AlCrFeCoNiCuTi alloy contains BCC1 phase, BCC2

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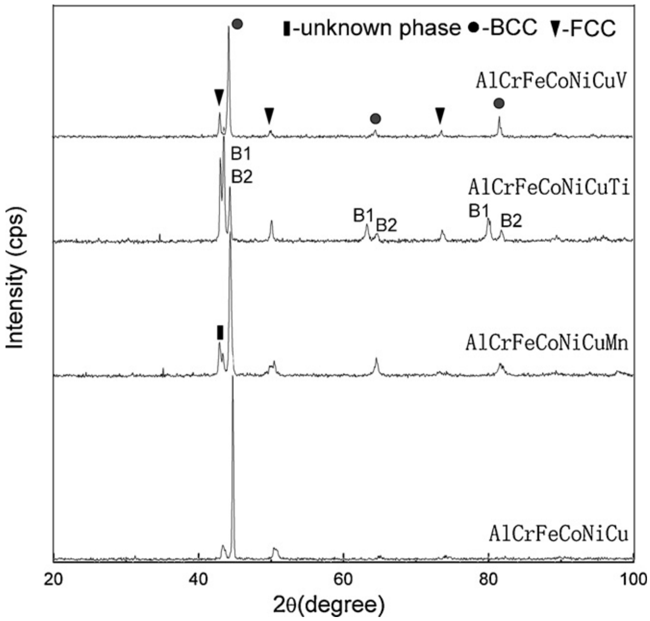

Fig. 1. XRD analyses of high entropy alloys.

phase and a FCC phase. AlCrFeCoNiCuV alloy contains two different phases with BCC, FCC structure, respectively. According to the relative intensity of diffractive peak, the main phases of these alloys are all BCC phases. Mn and V additions did not shown evident effect on the intensity of reflection peaks. While the additions of Ti increase the relative intensity of FCC phase. The reflection peaks of the BCC phases and FCC phases are shifted from each metal element due to intensive solid solution effect.

3.2. Microstructure

Fig. 2 shows the microstructure of the as-cast AlCrFeCoNiCu alloy. A dendrite structure was observed in the solidified AlCrFeCoNiCu alloy. The chemical compositions of the dendrite area (labelled A in Fig. 2(b)) and interdendrite area (labelled B in Fig. 2(b)) characterized by EDS are summarized in Table 1. It is shown that the dendrite area is rich in Cr, Fe, Co and interdendrite area is rich in Cu. Al and Ni elements are slightly rich in dendrite area. High-magnified image (Fig. 2(b)) presents that modulated plates consisted of nanoscale phases appear in the dendrite regions.

Fig. 3 shows the microstructure of the as-cast AlCrFeCoNiCuMn high entropy alloy. Three typical structures were observed in this alloy, labelled as A–C in Fig. 3(a). High-magnified image (Fig. 3(b)) shows similar modulated plates in zone A to that in the AlCrFeCoNiCu alloy. The composition in this zone A (shown in Table 1) are measured also similar to those in the AlCrFeCoNiCu alloy. Like the interdendrite region of AlCrFeCoNiCu alloy, the bone-shaped zone B is rich in Cu. The long-strip shape zone C is rich in Cr element. Consequently the Cr-enriched zone C is corresponding to the unknown phase shown in Fig. 1. Compared with the microstructure of AlCrFeCoNiCu alloy, Mn added to the AlCrFeCoNiCu induces the high melting point Cr-enriched phase first crystallized from the liquid alloy.

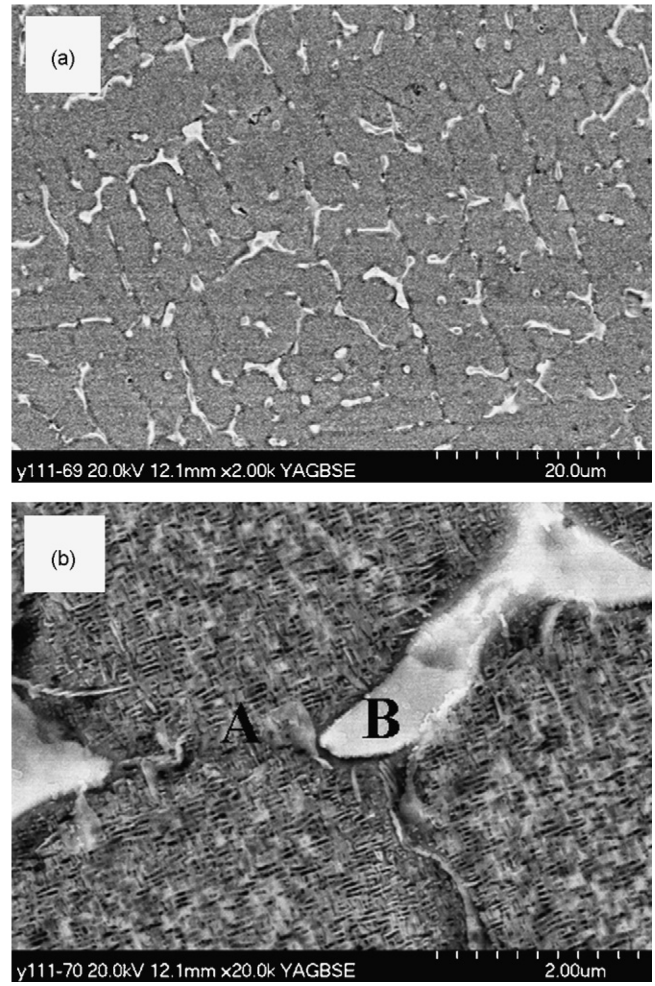

Fig. 2. SEM microstructure of AlCrFeCoNiCu alloy: (a) low-magnified image and (b) high-magnified image.

Fig. 4 shows that the as-cast AlCrFeCoNiCuTi alloy presents a two-phase eutectic-cell structure. The size of eutectic phases increases gradually from the eutectic cell cores to the cell walls because of the gradual release of crystal latent heat. The eutectic two-phases are labelled as A and B in Fig. 4(b) and phases between

Table 1
EDS analysis results of high entropy alloys

Sample region	AlCrFeCoNiCu		AlCrFeCoNiCuMn			AlCrFeCoNiCuTi			AlCrFeCoNiCuV	
	A	B	A	B	C	A	B	C	A	B
Al	14.42	9.93	16.93	9.06	1.24	20.35	6.3	5.21	10.01	10.27
Cr	19.66	3.12	10.58	3.37	79	5.52	37.5	2.22	17.73	2.55
Fe	18.37	4.80	15.85	5.38	8.06	8.69	24.26	4.57	17.13	3.17
Co	18.91	5.04	16.57	5.56	3.24	20.36	11.79	2.43	16.20	3.16
Ni	15.52	12.86	17.23	11.4	1.69	17.77	8.44	7.64	14.43	10.47
Cu	13.12	64.25	12.72	53.79	3.54	7.25	4.01	74.59	6.83	68.8
Mn	0	0	10.13	11.48	3.33	0	0	0	0	0
Ti	0	0	0	0	0	20.06	8.06	3.34	0	0
V	0	0	0	0	0	0	0	0	16.68	1.58

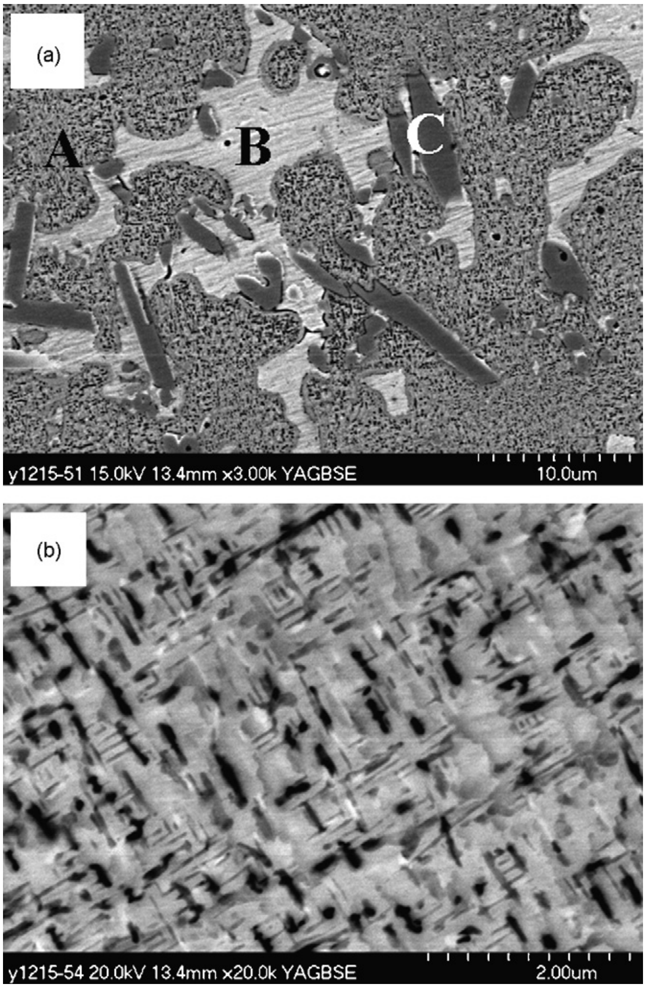

Fig. 3. SEM microstructure of AlCrFeCoNiCuMn alloy: (a) low-magnified image and (b) high-magnified image of zone A in Fig. 2(a).

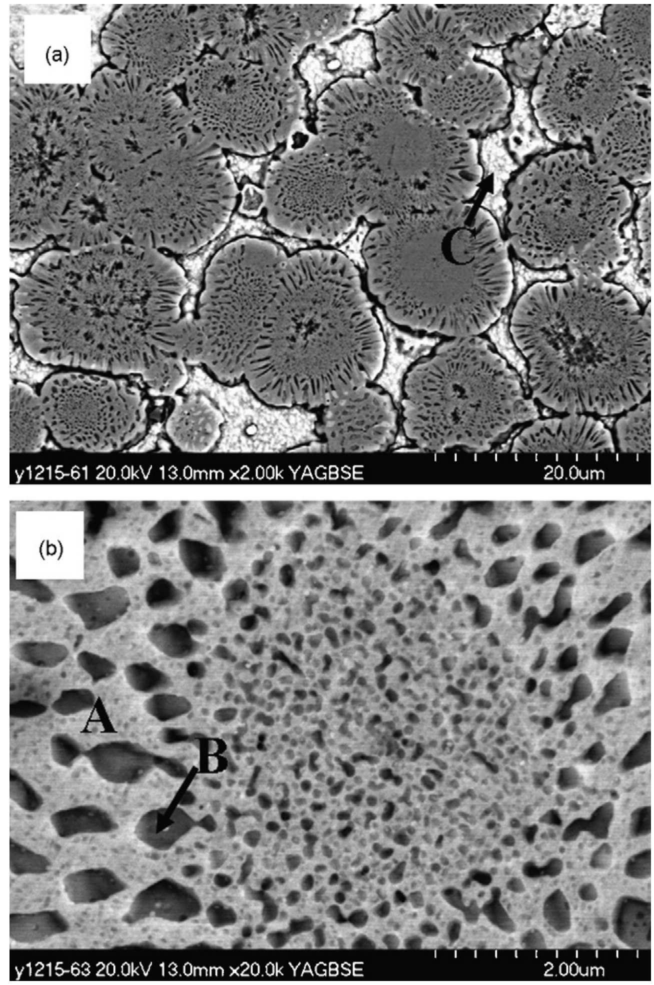

Fig. 4. SEM microstructure of AlCrFeCoNiCuTi alloy: (a) low-magnified image and (b) high-magnified of eutectic cell.

the eutectic cells are labelled as C in Fig. 4(a). EDS results (Table 1) indicate that phase A enriches in Al, Ti, Co and Ni, phase B enriches in Cr, Fe and phase C is a Cu rich region. Ti added to the AlCrFeCoNiCu alloy shows a significant effect on the microstructure. The addition of Ti made the alloy form a eutectic-cell structure. Based on the fact that almost all of the binary phase diagrams of Ti–M (M refer to Co, Cu, Fe and Ni) are eutectic type [20], so it may be reasonable to conceive a pseudo-binary eutectic structure in multi-element alloys.

Fig. 5(a) presents the TEM morphology of the eutectic cell. As shown in Fig. 5(b) and (c), the lattice constants of phase A and phase B are estimated to be 0.293 and 0.290 nm, respectively. It is known from Fig. 1 that BCC1 phase has a bigger interplanar distance indicating a larger lattice constant than BCC2. Thus, a conclusion can be made that the phase A is corresponding to BCC1 and the phase B is corresponding to BCC2.

The as-cast AlCrFeCoNiCuV is observed a typical dendrite structure as shown in Fig. 6. EDS results (Table 1) indicate that the interdendritic area is Cu rich region and dendritic area has similar concentrations for each element. However, one can also observe nano-sized precipitates within the dendritic area as shown in TEM bright image (Fig. 6(b)). Thus V additions change the morphology of dendritic area from modulated plates to ellipsoidal particles.

Single solid solution phase will maximize the role of mixing entropy, high entropy alloys are expected to yield single solid solution. However, the importance of high mixing entropy in sta-

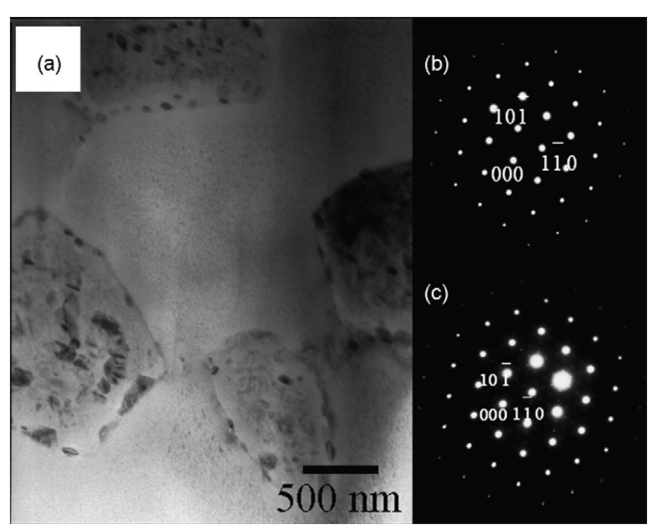

Fig. 5. TEM microstructure and SEAD pattern of eutectic cell of AlCrFeCoNiCuTi alloy: (a) bright field TEM image; (b) SAED patterns of phase A along [1 1 1] zone axis; (c) SAED pattern of phase B along [1 1 1] zone axis.

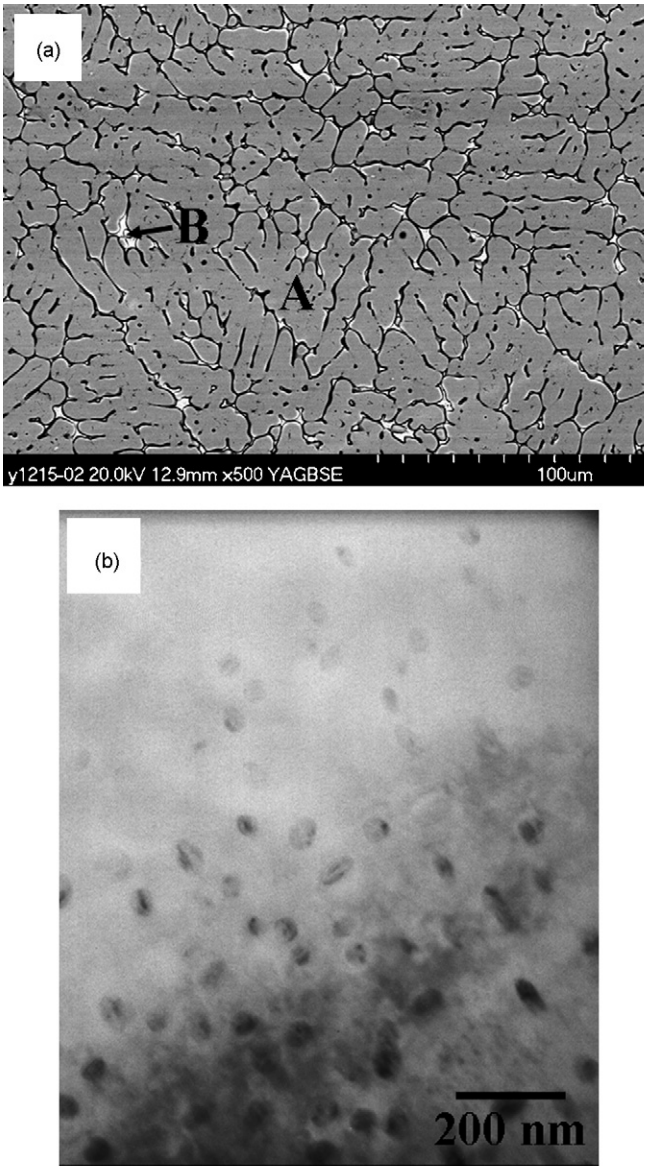

Fig. 6. Microstructure of AlCrFeCoNiCuV alloy: (a) SEM image and (b) TEM image of zone A in Fig. 4(a).

bilizing solid solutions is reduced with the decreasing temperature according to the Gibbs formula $\Delta G = \Delta H - T\Delta S$. Element like Cu which have positive mixing enthalpy or small negative mixing enthalpy with other metals is rejected to the grain boundary during solidification and formed Cu enriched phase. From the EDS results (Table 1), it can be seen the additions of Mn to AlCrFeCoNiCu alloy decrease the Cu segregation. Mn was also found to inhibited Cu segregation in other high entropy alloys [21]. While the additions of Ti and V were found to promote the segregation of Cu.

3.3. Compressive properties

Fig. 7 presents the stress–strain curves of these high entropy alloys under compression at room temperature. The yield strength, compressive strength and ultimate strain are listed in Table 2. It can be seen that the yield stress of AlCrFeCoNiCu alloy is 1303 MPa. Due to a large strain hardening ability, the compressive strength reaches 2018 MPa and an ultimate strain of 24%.

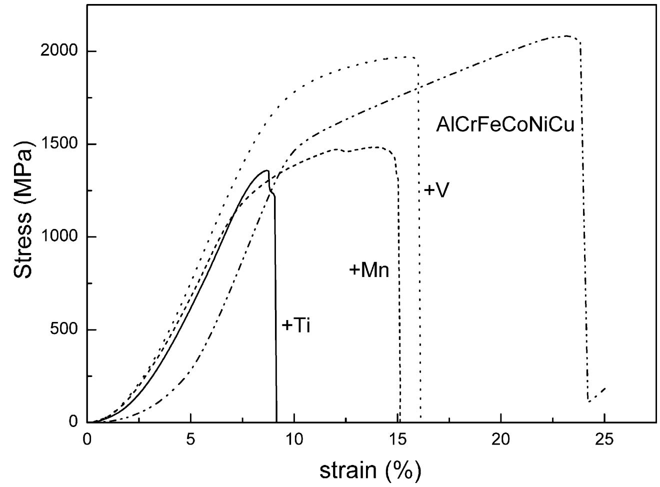

Fig. 7. Stress–strain curves of high entropy alloys under compression.

The additions of Mn, Ti and V by equal molar ratio can decrease the compressive strength and ultimate strain of the AlCrFeCoNiCu. AlCrFeCoNiCuMn alloy contains long-strip shape Cr-enriched phase which must have significant influence on mechanical properties, especially at the crack nucleation stage. Because the nucleation of crack is a highly localized process that occurs at various microstructural inhomogeneties.

AlCrFeCoNiCuTi alloy shows almost no strain hardening ability, thus its compressive strength and ultimate strain are the lowest. The recent work reported by Zhang et al. [22] has also shown that when the content of Ti is equimolar with other element in high entropy alloys, the plastic strain decreases rapidly. A small strain hardening ability was observed of AlCrFeCoNiCuMn and AlCrFeCoNiCuV alloy. Among the three modified alloys, the AlCrFeCoNiCuV alloy shows a highest strength and largest ductility. The compressive strength of AlCrFeCoNiCuV alloy is reach to 1970 MPa and the ultimate strain is 16%. AlCrFeCoNiCuV alloy has lager strength and ductility than Mn, Ti addition alloys, but still lower than AlCrFeCoNiCu alloy. This may be due to the modulated plate structure in dendrite region of AlCrFeCoNiCu alloy. The modulated plate structure was consisted of two-phase with the same lattice constant [17]. The compatible deformation of the two phases is responsible for large strain hardening ability.

Mn and Ti add to AlCrFeCoNiCu alloy also show a decrease of yield strength. While the yield strength of AlCrFeCoNiCuV alloy is 1469 MPa which is higher than that of AlCrFeCoNiCu alloy manifesting that V addition aid to increase the yield strength. Therefore, one can know that the addition of V into the AlCrFeCoNiCu alloy shows the best strengthening effect and the lowest decrease in the ultimate strain in our experiments. This may because V addition did not cause extra phase present. Except Cu segregation in interdendrite, massive elements of different atomic sizes solute in dendrite region. For substitutional solid solution, heterogeneous atoms with different atomic sizes combined together will lead to lattice distortion. Simultaneous the density of dislocation increases.

Table 2
Room temperature compressive properties of high entropy alloys

Sample	σ_y (MPa)	σ_b (MPa)	ε (%)
AlCrFeCoNiCu	1303	2081	24
AlCrFeCoNiCuMn	1005	1480	15
AlCrFeCoNiCuTi	1234	1356	9
AlCrFeCoNiCuV	1469	1970	16

Stress field induce by distortion interacted with elastic stress field around dislocation, made atoms aggregated around dislocation line form dislocation atmosphere. Dislocation slip must overcome the pinning of atmosphere, together with atmosphere or break free from atmosphere, so that the dislocation slip shear stress increases. Strong binding and high melting point elements, such as V, are more efficient in increasing the slip resistance.

4. Conclusions

Additions of equal-molar Mn, Ti and V can change the microstructure and mechanical properties of a high entropy AlCrFeCoNiCu alloy. The AlCrFeCoNiCuMn alloy shows the same microstructure as the AlCrFeCoNiCu alloy except for a long-strip shape Cr-enriched phase. Ti added to the AlCrFeCoNiCu alloy changes its microstructure from the dendrite to a eutectic cell. The eutectic cell includes one Al, Ti, Co, Ni enriched phase and another Cr, Fe enriched phase. Between the eutectic cells, there is a Cu enrich phase. The additions of Ti can increase the relative intensity of FCC phase. AlCrFeCoNiCuV alloy shows also a dendrite structure but there distribute quantities of nano-scale particles homogeneously in the dendrite area.

Mn, Ti, V added to AlCrFeCoNiCu by equal molar ratio decreases the compressive strength and ductility. The long-strip shape Cr-enriched phase may induce the decreases of strength and plasticity of the AlCrFeCoNiCuMn alloy. The AlCrFeCoNiCuTi alloy has almost no strain hardening ability, its compressive strength and ultimate strain is the lowest. Among the three modified alloys, the AlCrFeCoNiCuV alloy shows a highest strength and largest ductility. The compressive strength of the AlCrFeCoNiCuV alloy reaches to 1970 MPa and the ultimate strain is 16%. The yield strength of the AlCrFeCoNiCuV alloy is higher than that of AlCrFeCoNiCu alloy. Therefore, the addition of V into the AlCrFeCoNiCu alloy shows the best strengthening effect and the lowest decrease in the ultimate strain in our experiments.

References

[1] J.W. Yeh, Y.L. Chen, S.J. Lin, S.K. Chen, Mater. Sci. Forum. 560 (2007) 1–9.
[2] Handbook Committee, Metals Handbook, vol. 1, 10th ed., ASM International, Metal Park, OH, 1990, pp. 3–949.
[3] Handbook Committee, Metals Handbook, vol. 2, 10th ed., ASM International, Metal Park, OH, 1990, pp. 3–757.
[4] A.L. Greer, in: J.H. Westbrook, R.L. Fleischer (Eds.), Intermetallic Compounds—Principles and Practice, vol. 2, Wiley, New York, 1995, pp. 53–209.
[5] A.L. Greer, Nature 366 (1993) 303–304.
[6] S. Ranganathan, Curr. Sci. 85 (2003) 1404–1406.
[7] J.W. Yeh, S.K. Chen, S.J. Lin, J.Y. Gan, T.T. Shun, C.H. Tsau, S.Y. Tsau, Adv. Eng. Mater. 6 (2004) 299–303.
[8] J.W. Yeh, S.K. Chen, J.Y. Gan, S.J. Lin, T.S. Chin, T.T. Shun, C.H. Tsau, S.Y. Chang, Metall. Mater. Trans. A 35A (2004) 2533–2536.
[9] C.C. Tung, J.W. Yeh, T.T. Shun, S.K. Chen, Y.S. Huang, H.C. Cheng, Mater. Lett. 61 (2007) 1–5.
[10] Y.J. Zhou, Y. Zhang, Y.L. Wang, G.L. Chen, Appl. Phys. Lett. 90 (2007) 1–3.
[11] W.H. Wu, C.C. Yang, C.C.J.W. Yeh, Ann. Chim. Sci. Mat. 31 (2006) 737–747.
[12] Y.Y. Chen, T. Duval, U.D. Hung, J.W. Yeh, H.C. Shih, Corros. Sci. 47 (2005) 2257–2279.
[13] J.M. Wu, S.J. Lin, J.W. Yeh, S.K. Chen, Y.S. Huang, H.C. Cheng, Wear 261 (2006) 513–519.
[14] P.K. Hung, J.W. Yeh, T.T. Shun, S.K. Chen, Adv. Eng. Mater. 6 (2004) 74–78.
[15] T.K. Chen, T.T. Shun, J.W. Yeh, M.S. Wong, Surf. Coat. Technol. 188–189 (2004) 193–200.
[16] Y.Y. Chen, U.T. Hong, J.W. Yeh, H.C. Shih, Scripta Metall. 54 (2006) 1997–2001.
[17] C.J. Tong, Y.L. Chen, S.K. Chen, J.W. Yeh, T.T. Shun, C.H. Tsau, S.J. Lin, S.Y. Chang, Metall. Mater. Trans. A 36A (2005) 881–893.
[18] C.J. Tong, M.R. Chen, S.K. Chen, J.W. Yeh, T.T. Shun, S.J. Lin, S.Y. Chang, Metall. Trans. A 36A (2005) 1263–1271.
[19] G.Y. Ke, C.W. Tsai, C.C. Tung, J.W. Yeh, T.T. Shun, C.C. Yang, S.W. Chen, Ann. Chim. Sci. Mat. 31 (2006) 685–698.
[20] Handbook Committee, Metals Handbook, vol. 3, 10th ed., ASM International, Metal Park, OH, 1990, pp. 2150–2319.
[21] H.Y. Chen, S.K. Chen, T. Hsu, J.W. Yeh, Chen, Ann. Chim. Sci. Mat. 31 (2006) 669–683.
[22] X.F. Wang, Y. Zhang, Y. Qiao, G.L. Chen, Intermetallics 15 (2007) 357–362.